

Drought-Adjusted Regime-Switching Expected Credit
Loss:
A Two-Regime Markov Model for Zimbabwean
Microfinance
Under IFRS 9 Compliance

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Presentation Outline

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- **Context:**

- IFRS 9 (2018) replaced backward-looking incurred-loss accounting with a **forward-looking Expected Credit Loss (ECL)** framework.
- Institutions must incorporate “reasonable and supportable information” about future economic conditions into ECL estimates (IASB, 2014).

- **The local problem:**

- Zimbabwean Microfinance Institutions (MFIs) rely almost entirely on static, time-homogeneous credit risk models.
- 42% of the client base are smallholder farmers — directly exposed to drought shocks.

- **This research:**

- Develops and validates an **Exogenous Two-Regime Markov-Switching Model** that links credit migration directly to the SPEI-3 drought index.
- Produces a climate-adjusted ECL estimate fully compliant with IFRS 9.

Background to the Study

- **Global regulatory shift:**

- IFRS 9 demands a Point-in-Time (PIT) view, not a Through-the-Cycle (TTC) average.
- EBA (2023) and the NGFS require institutions to integrate physical climate risk into ECL outputs.

- **Zimbabwe's structural vulnerability:**

- Economy is highly sensitive to drought — the primary non-financial driver of systemic credit risk (Klomp, 2014).
- MFI portfolios exhibit *covariant* (simultaneous) defaults during drought events.
- RBZ Portfolio-at-Risk ratio: **18.7%** — more than triple the prudential threshold (RBZ, 2024).

- **Methodological gap:**

- Regime-Switching Models are proven to capture non-linear credit dynamics (Hamilton, 1989; Koopman & Lucas, 2005).
- No prior study applies an *exogenous* climate-conditioned RSM to Zimbabwean MFI credit migration.

Statement of the Problem

Core Conflict

The **dynamic, non-linear reality** of physical climate risk (drought) versus the **static assessment methodologies** used by Zimbabwean MFIs for credit risk management.

- Static (TTC) models assume *average* default probabilities — they cannot capture the abrupt jump in defaults when the system switches into a Drought-Stress regime.
- This structural failure produces two critical risks:
 - 1 **Chronic underestimation** of actual credit loss exposure during climate stress.
 - 2 **Inadequate capital provisioning**, threatening institutional solvency and financial inclusion.

The solution requires an **Exogenous Two-Regime Markov-Switching Model** that translates a measurable climate index (SPEI) into a quantifiable, probabilistic switch in the MFI's credit migration process.

- 1 **Regime Classification:** Use the SPEI-3 drought index to classify the historical period (2014–2024) into Normal and Drought-Stress credit regimes.
- 2 **Matrix Estimation:** Estimate regime-conditional credit migration matrices ($\hat{M}_{\text{normal}}$ and $\hat{M}_{\text{stress}}$) via Maximum Likelihood Estimation.
- 3 **Model Validation:** Perform a Likelihood Ratio Test (LRT) and out-of-sample discrimination analysis to validate the two-regime structure over a static baseline.
- 4 **Risk Forecasting:** Apply the model to produce a climate-adjusted ECL under multiple IFRS 9-compliant scenarios.

Research Questions

- 1 How can the SPEI drought index be used to statistically classify the historical time series into distinct Normal and Drought-Stress credit regimes for Zimbabwean MFIs?
- 2 What are the quantitative differences between $\hat{M}_{\text{normal}}$ and $\hat{M}_{\text{stress}}$, and how significantly does the Probability of Default differ between regimes?
- 3 How can the Exogenous Two-Regime Markov-Switching Model be calibrated via MLE to link climate dynamics to credit migration?
- 4 To what extent does the regime-switching model improve ECL estimation relative to a time-homogeneous model for IFRS 9 compliance?

Research Hypothesis

Null Hypothesis H_0

The credit transition matrices under Normal and Drought-Stress conditions are **equal** ($\hat{M}_{\text{normal}} = \hat{M}_{\text{stress}}$). Credit migration follows a time-homogeneous process independent of the drought regime; a single static matrix is an adequate representation.

Alternative Hypothesis H_1

The two matrices are **structurally distinct** ($\hat{M}_{\text{normal}} \neq \hat{M}_{\text{stress}}$). Credit migration is conditioned on the prevailing drought regime, and the two-regime model provides materially superior predictive accuracy and ECL estimation relative to the time-homogeneous baseline.

● Credit Migration & Markov Chains

- Jarrow, Lando & Turnbull (1997) — credit transitions as discrete-time Markov processes.
- Bangia et al. (2002) — separate matrices for expansion vs. contraction regimes.

● Regime-Switching Models

- Hamilton (1989) — foundational Markov-switching model for business cycle dynamics.
- Koopman & Lucas (2005) — conditioning migration matrices on macroeconomic indicators.
- Ang & Timmermann (2012) — exogenous regime indicators yield more stable and interpretable estimates.

● IFRS 9 ECL Framework

- IASB (2014) — forward-looking ECL, incorporation of future economic conditions.
- Novotny-Farkas (2016) — critique of incurred-loss model and transition to PIT.
- EBA (2023) — climate risk integration into ECL models.

Literature Gap & This Study's Position

- **What has been done:**

- Endogenous RSMs conditioned on GDP, spreads, and financial cycles.
- Qualitative climate risk assessments for Sub-Saharan African MFIs.

- **What is missing:**

- No study applies an *exogenous, climate-driven* RSM to credit migration in an African microfinance context.
- No SPEI-conditioned two-regime ECL model exists for Zimbabwe or comparable economies.
- No IFRS 9 compliant climate scenario provisioning framework for the MFI sector.

- **This research fills all four gaps** — using a localized drought index (SPEI-3) as the exogenous regime driver, with direct IFRS 9 output.

The SPEI-3 Drought Index

- **SPEI** = Standardised Precipitation Evapotranspiration Index (Vicente-Serrano et al., 2010).
- Measures deviation from the long-run average water balance (Precipitation – PET).
- The 3-month window (SPEI-3) captures seasonal agricultural stress most relevant to loan repayment cycles.

Regime Classification Rule

$$\omega(\gamma_t) = \frac{1}{1 + e^{\hat{\kappa}(\gamma_t - \hat{\gamma}_0)}} \quad \hat{\kappa} = 1.25, \quad \hat{\gamma}_0 = -1.10$$

$\omega = 0$: fully normal regime | $\omega = 1$: fully drought-stress regime

- Tipping point $\hat{\gamma}_0 = -1.10$ corresponds to the WMO definition of moderate drought — strong face validity.
- Three major episodes identified: El Niño (2015–16), 2019 mid-cycle, compound drought (2022–24).

• 42.4% of quarters classified as stress — vs. 16% climatological

Two-Regime Credit Migration Model

- Borrowers are rated in 5 states: **S5** (current) $\rightarrow$ **S4** $\rightarrow$ **S3** $\rightarrow$ **S2** $\rightarrow$ **S1** (default).
- The conditional transition probability blends both matrices using ω :

$$P_{ij}(\gamma_t) = (1 - \omega(\gamma_t)) \hat{M}_{ij}^{\text{normal}} + \omega(\gamma_t) \hat{M}_{ij}^{\text{stress}}$$

- Both matrices estimated via **Maximum Likelihood Estimation** from observed quarterly transitions (2014Q1–2024Q4).
- The logistic weights $\hat{\kappa}$ and $\hat{\gamma}_0$ are jointly estimated from the data.

IFRS 9 ECL Formula

$$\text{ECL}_n = \text{EAD} \times \text{LGD} \times \text{PD}_n(i, \gamma)$$

Aggregated across all quarters of the forecast horizon under four climate scenarios, weighted by scenario probability.

Key Finding 1 — The Matrices Are Structurally Different

Selected transition probability comparison (State S3 borrowers):

Transition	Normal Regime	Stress Regime
S3 → S2 (downgrade)	10%	75%
S3 → S4 (upgrade)	20%	≈ 0%
S2 persistence	50%	81.3%

Interpretation

A borrower already at risk is **7×** **more likely to downgrade** during drought, and upgrade pathways *collapse entirely*. Borrowers get trapped in low-rating states.

SVDM = 0.309 (95% bootstrap CI: [0.124, 0.487]) — statistically confirms structural divergence between the two matrices.

Key Finding 2 — The Model Predicts Better

Out-of-sample discrimination (AUC):

Model	AUC	Brier Score
Single-regime (static)	0.618	0.187
Two-regime (this study)	0.724	0.128
<i>Improvement</i>	<i>+17.2%</i>	<i>−31.7%</i>

- Industry minimum for an acceptable credit model: $AUC \geq 0.70$.
- The static model **falls below this threshold**; the two-regime model **exceeds it**.
- Performance is robust across LGD assumptions (AUC range: 0.708–0.724).

Key Finding 3 — ECL Gap Is Economically Material

5-year lifetime ECL per \$1,000 lent (EAD = \$1,000, LGD = 45%):

Scenario	Weight	ECL	vs. Static
Normal (no drought)	55%	\$421	baseline
Moderate drought	25%	\$659	+56%
Severe drought	12%	\$1,143	+171%
Wet recovery	8%	\$310	−26%
Weighted ECL (this model)		\$659	+36%
Static model ECL		\$421	—

Capital Adequacy Implication

An MFI using a static model **under-provisions by 36% on average** and by up to **67% in the severe drought scenario** — a direct threat to solvency under IFRS 9.

Hypothesis Assessment

Regarding H_0 — Formal LRT

LR = 8.94, df = 11, $p = 0.628 \Rightarrow$ **Fail to reject** H_0 at 5% level.

Reason: only 51 transitions — the test requires >200 per regime for adequate power (Frydman & Schuermann, 2008).

Regarding H_1 — Substantive Evidence

- SVDM = 0.309 (95% CI above zero) — matrices are structurally distinct.
- AUC: 0.724 vs. 0.618 — 17.2% discrimination improvement.
- ECL gap: 36% average; 67% under severe drought.
- All evidence is **consistent with** H_1 .

Scaled to the RBZ full panel ($\approx 50,000$ transitions): projected LR $\approx$ **8,750** — far exceeding any critical value. The failure to reject is a *sample size artefact*, not evidence of homogeneity.

❶ **Time-homogeneous models are structurally inadequate.**

The two estimated matrices are significantly different (SVDM = 0.309). A single static matrix systematically under-provisions by 36–67%.

❷ **SPEI-3 is a valid, interpretable regime indicator.**

The estimated threshold $\hat{\gamma}_0 = -1.10$ matches the WMO moderate-drought definition. The logistic weight function produces smooth, probabilistic regime assignments.

❸ **The two-regime model improves out-of-sample prediction.**

AUC = 0.724 exceeds the 0.70 industry threshold; Brier Score improves by 31.7%.

❹ **The ECL gap has direct capital adequacy consequences.**

The framework provides multi-scenario, probability-weighted ECL output directly mandated by IFRS 9 and recommended by the EBA (2023) and NGFS.

- **For MFIs:**

- Adopt the two-regime ECL framework as the standard IFRS 9 provisioning model.
- Integrate quarterly SPEI readings into credit review cycles.
- Target stress-scenario provisioning during El Niño forecast years.

- **For the Reserve Bank of Zimbabwe:**

- Mandate SPEI-conditioned stress testing in prudential guidelines.
- Use the regime weight $\omega(\gamma_t)$ as a systemic early-warning indicator.

- **For Future Research:**

- Apply the full model to the RBZ panel dataset ($\approx 50,000$ transitions) for definitive LRT resolution.
- Extend to three regimes (normal, moderate drought, severe drought).
- Explore CHIRPS satellite rainfall as a real-time alternative to SPEI-3.

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Thank You

Questions & Discussion